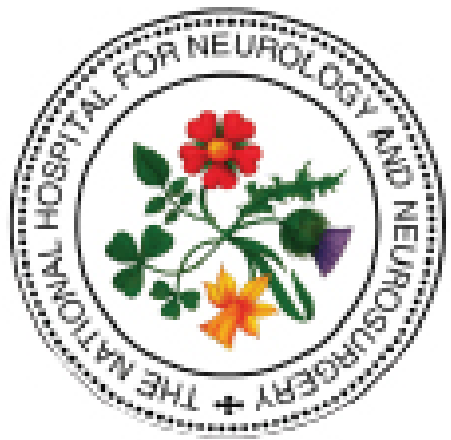
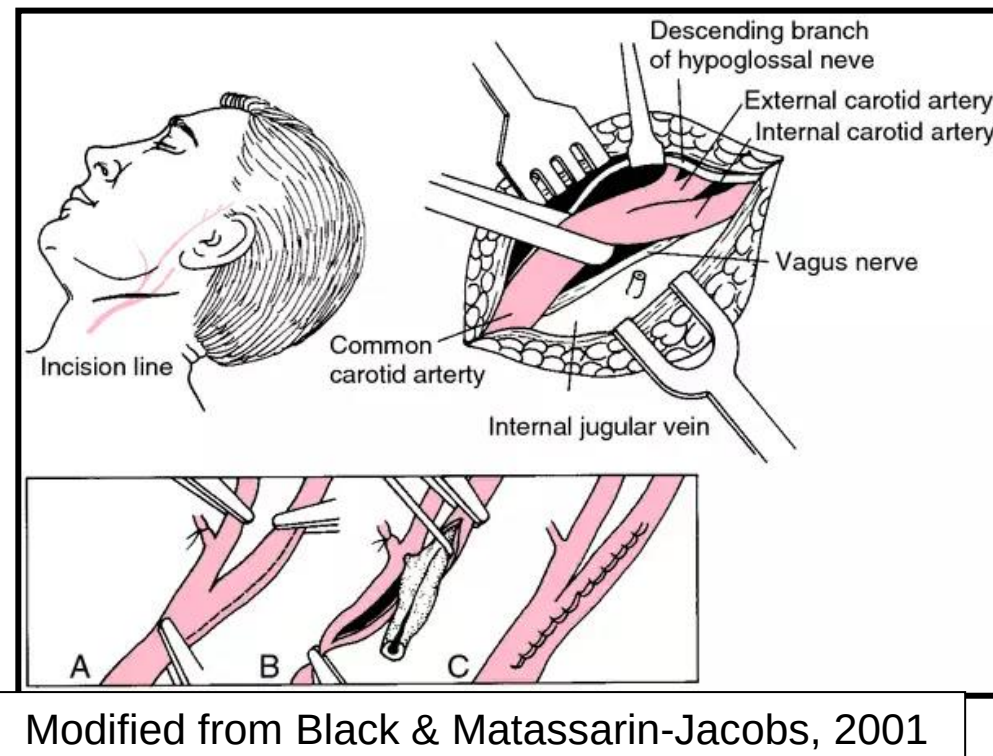




Carotid Endarterectomy (CEA)

Postoperative HDU Care / **WARNING** Signs



CEA - surgical procedure used to correct carotid stenosis (narrowing of carotid artery lumen by atheroma) – surgery can reduce risk of transient ischemic attacks (TIAs) or stroke.

ENDARECTOMY - removal of plaque from artery walls.

ANGIOPLASTY / STENTING of carotid artery - alternative treatment to CEA.

Signs to look out for post-CEA:

CAROTID ENDARTERECTOMY	
Post-operative signs and symptoms which are a WARNING of possible complications and require immediate REVIEW	
W	Wound / neck swelling / bleeding
A	Airway compromise / breathing difficulties / tachypnoea / hoarseness / voice changes / dysphagia
R	Rhythm changes / heart rate changes
N	Neurological deterioration / confusion / seizure
I	Increasing / decreasing BP outside specified post-operative parameters
N	New or increasing headache unresolved with analgesia
G	Get immediate REVIEW by medical team if any of above signs or symptoms present
MEDICAL DETERIORATION	Stroke team (bleep 1146/8158) + UCH Vascular team (bleep 5026) + Outreach (bleep 8277) + Anaesthetic SpR (1 st on call bleep 8131)
WOUND COMPLICATIONS	NHNN NV Surgical team (bleep 2137 / 8244) + UCH Vascular team (bleep 5026) + Outreach (bleep 8277) + Anaesthetic SpR (1 st on call bleep 8131)

CEA patients attached to **Sedline 03 Regional Oximetry** to monitor cerebral oxygenation intra-op (helps determine need for stenting)

Monitoring will be continued post-op in HDU but does not need recording by nursing staff

Data will subsequently be retrieved from the Sedline monitor by medical staff

NEUROLOGY

GCS, Pupils, Limb assessment - as per *Postoperative Observations Protocol*

Assess cranial nerves

- **HYPOGLOSSAL** - ability to position tongue midline - *if deviates call surgeon* - assess hypoglossal pharyngeal nerve for swallowing ability
- **FACIAL** - Ask patient to smile and puff cheeks
- **VAGUS** - Ask patient to speak, if unable to speak or speech garbled, call surgeon
- **SPINAL ACCESSORY** – Ask patient to bilaterally raise arms and sustain for 3sec

If patient has **N**eurological deterioration / confusion or symptoms of severe headache / seizure / uncontrolled hypertension – *this is an emergency and must be escalated to medical staff immediately to control hypertension with IV labetalol & start phenytoin for seizure control*

VITAL SIGNS

Manipulation of carotid bulb during CEA can result in haemodynamic instability in early postoperative period

Heart rate

Monitor **HR** – report any **R**hythm / **R**ate changes
Continuous ECG monitoring

Blood pressure

Maintain **BP** within specified postoperative parameters

Hypotension – *can cause low cerebral blood flow and cerebral ischaemia*

Hypertension - *may increase risk of neck haematoma or suture line disruption*

Respiratory

Observe for **A**irway compromise / breathing difficulties / tachypnoea / hoarseness / voice changes / dysphagia

WOUND

Observe for **W**ound / neck swelling / bleeding / airway compromise (as per **WARNING** signs)

- Monitor wound drainage
- NCC Nursing staff to remove drain according to Vascular Team instructions

1. **RELEASE SUCTION** on wound drain with **AIR INLET NEEDLE**
2. Remove drain **SLOWLY** to avoid **VASOVAGAL STIMULATION**
3. Once removed, dispose of air inlet in sharps bin and drain and tubing in clinical waste bin

Systolic BP

>180mmHg

OR

>140mmHg

+ any of following symptoms
Headache / GCS <15 / Seizure

>150mmHg

100-150mmHg

Action(s)

1. IV Labetalol, 10mg every 10min until SBP <160mmHg
2. Aim <140mmHg if symptomatic

1. Ensure regular oral antihypertensives restarted
2. Consider hydralazine and amlodipine if HR <40
3. Symptomatic hypertension may require imaging - liaise with Vascular surgeons (bleep 5026)

1. Administer regular antihypertensive drugs and consider higher or repeated doses
2. Consider additional oral antihypertensive
3. Consider IV labetalol 10mg every 10 min if refractory
4. Review regular antihypertensive medications

Observe / Continue regular antihypertensives

IV antihypertensive therapy for immediate control

1st line IV labetalol as bolus (up to 100mg max)

2nd line IV hydralazine 2mg every 5 min (up to 10mg max)

3rd line GTN as infusion – start at 0.5mg/h (range 0.5 -12mg/h)

Oral antihypertensive medications

- Usual antihypertensives taken on morning of surgery, with exception of angiotensin-converting enzyme (ACE) inhibitors and angiotensin II receptor blockers (ARBs)
- **All** regular antihypertensives restarted postop, only omitted if hypotensive
- May need higher and / or repeated doses of regular medications
- May require additional oral antihypertensive drugs

Additional oral antihypertensive therapy

Patient currently taking	Consider adding
A	C (Amlodipine 5mg)
C	A (Ramipril 5mg)
D	A (Ramipril 5mg)
A + C	D (Bendroflumethiazide 2.5mg)
A + D	C (Amlodipine 5mg)
A + C + D	B (Bisoprolol 5mg)

A = ACE inhibitor, **B** = Beta blocker,

C = Calcium channel blocker, **D** = Diuretic

Avoid Nifedipine as can cause rapid and unpredictable drop in BP

Angiotensin II Receptor Blocker Examples

Suffix “**sartan**”

Losartan

Valsartan

Candesartan

Eprosartan

Olmesartan

